

CS 401R: Engineering Production AI Systems

Brigham Young University – Fall 2026

Instructor: Scott Toborg

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Meeting Times: Tuesday & Thursday, 2:00-3:15 pm

First Day of Class: Thursday, September 3, 2026

Last Day of Class: Thursday, December 10, 2026

Credits: 3

Format: In-person lecture

Course Description

This course teaches engineers how to build, ship, and operate AI systems at production scale inside real enterprises. We move from theory to working systems: you will design platform architectures, build data and model pipelines, rigorously evaluate AI outputs, deploy with confidence, and then operate those systems with the economic, governance, and reliability discipline that enterprise stakeholders demand.

The course is organized around a single running case study – **NorthStar Retail** – a fictional but architecturally realistic enterprise AI deployment. Every lab builds one layer of that system. By the end of the semester, you will have designed and prototyped a complete, end-to-end enterprise AI platform on AWS.

The highlight of the course is the team project. You will organize into teams of 3-4 students and build an end-to-end AI System of your choice. This is where you can think big and create something you'd be proud of showing off to potential employers, investors, graduate search committees, etc. Some past student projects have led to full commercial deployments and inspiration for follow-on start-ups.

Primary text: *Engineering the AI Enterprise: Orchestrating Strategy, Product, and Execution* (Toborg, 2026) – Parts 3 (Build) and 4 (Operate). Draft chapters will be distributed via Canvas. Please do not share outside the course – this is pre-publication copyrighted material.

Prerequisites

- CS 240 or equivalent (Advanced Software Construction)
- CS 270 or equivalent (Introduction to Machine Learning)
- Recommended courses: CS 301R (Agentic Systems), CS 329 (QA & DevOps), CS 452 (Data Engineering), CS 574 (Transformers)
- Strong experience in Python and working knowledge of SQL
- Familiarity with cloud computing concepts (AWS experience helpful but not required)

Graduate students without the undergraduate prerequisites may seek permission from the instructor.

Learning Objectives

By the end of this course, students will be able to:

1. Design a production-grade AI platform architecture on AWS, including infrastructure as code, feature stores, and model registries
2. Build end-to-end data and feature engineering pipelines that handle real-world distribution shift
3. Train, fine-tune, and deploy models across the full development spectrum – from prompt engineering to RAG to custom training to agent orchestration
4. Apply XOps discipline (DataOps, MLOps, LLMOps, AgentOps) to automate the model lifecycle
5. Implement CI/CD pipelines for AI systems with appropriate deployment strategies (canary, blue/green, shadow)
6. Evaluate AI system quality rigorously across predictive models, generative systems, and autonomous agents
7. Operate AI systems in production: monitoring, drift detection, reliability engineering, and incident response
8. Measure and communicate AI business value to both engineering and executive audiences
9. Manage AI costs using FinOps discipline
10. Design governance frameworks that scale to agentic AI systems

Required Materials and Costs

Primary Text: *Engineering the AI Enterprise* draft chapters (Parts 3 & 4), distributed as PDFs on Canvas.

AWS Environment: Students work in AWS SageMaker for all labs. Students will be enrolled in AWS Academy to get access to some free services. Students should also sign up for a free-tier account—see Canvas for setup instructions.

No other textbook required. Supplementary references are embedded in each chapter and linked on Canvas.

Total Costs: While we will do everything to minimize costs, there will be some unavoidable AWS charges estimated at around \$50. You will be responsible for all charges. If you rack up unintended charges, AWS is usually very forgiving. You will need a credit card to sign up for an AWS account.

Grading

Component	Weight	Notes
Labs (7 total)	49%	7.0% each (evenly weighted)
Final Project	30%	Team-based

Component	Weight	Notes
AWS Academy	10%	Foundation and GenAI courses
Quizzes	11%	In-class
Total	100%	

Late Policy: Labs lose 10% per day late. Contact me *before* the deadline if you have a documented emergency – not after.

Extra Credit: TBD up to 5% of grade.

Grade Scale: A 93+, A- 90-92, B+ 87-89, B 83-86, B- 80-82, C+ 77-79, C 73-76, below 73 see instructor.

Lab Overview – NorthStar Retail Arc

All 7 labs use the **NorthStar Retail** case study. Each lab is self-contained but contributes to a complete AI platform. Labs are assigned in class (Thursdays) and due **Saturday at midnight**, approximately two weeks after assignment.

Lab	Topic	Assigned	Due
Lab 1	Platform Foundation	Thu Sep 3	Sat Sep 19
Lab 2	Data & Feature Engineering	Thu Sep 17	Sat Oct 3
Lab 3	Model Development	Thu Oct 1	Sat Oct 17
Lab 4	CI/CD Pipeline	Thu Oct 15	Sat Oct 31
Lab 5	Deployment & Scaling	Thu Oct 29	Sat Nov 14
Lab 6	Monitoring & Reliability	Thu Nov 12	Sat Nov 28
Lab 7	Metrics, Economics & Business Value	Thu Nov 19	Tue Dec 1 (!)

(!) Lab 7 is due **Tuesday Dec 1 at midnight** – an exception to the Saturday convention given the Dec 10 end-of-semester constraint. All remaining class sessions after Dec 1 are devoted to the team project.

Lab Descriptions

Lab 1 – Platform Foundation Stand up the NorthStar AWS platform skeleton: SageMaker domain, S3 bucket structure, IAM roles, VPC configuration, and Terraform IaC scripts. Deliverable: a reproducible platform deployment with a written architecture decision record (ADR).

Lab 2 – Data & Feature Engineering Build NorthStar’s customer data pipeline: raw ingestion from simulated sources, Glue transformation jobs, and a SageMaker Feature Store with at least three engineered features. Deliverable: working pipeline + data lineage diagram.

Lab 3 – Model Development Choose one: (a) fine-tune a foundation model on NorthStar product data, or (b) implement a RAG system for NorthStar’s customer support assistant. Both op-

tions require a prompt evaluation harness. Deliverable: trained/deployed model + evaluation report.

Lab 4 – CI/CD Pipeline Implement a CI/CD pipeline using AWS CodePipeline that runs automated tests, promotes a model through staging, and registers it in the SageMaker Model Registry with appropriate stage gates. Deliverable: working pipeline + pipeline health dashboard.

Lab 5 – Deployment & Scaling Deploy the NorthStar recommendation model to a SageMaker endpoint using canary deployment. Configure auto-scaling policies and a rollback trigger. Deliverable: deployed endpoint + load test results + rollback runbook.

Lab 6 – Monitoring & Reliability Implement CloudWatch dashboards for NorthStar covering data drift, model performance degradation, and latency. Write an incident response runbook for three failure scenarios. Deliverable: dashboards + runbook + one simulated incident walk-through.

Lab 7 – Metrics, Economics & Business Value Instrument the NorthStar system end-to-end: build the Metric Pyramid linking model outputs to business KPIs, produce a FinOps cost attribution report (cost per recommendation, total platform spend, build-vs-buy comparison), and draft a shared scorecard bridging engineering and executive views. Deliverable: written report + supporting dashboard or spreadsheet artifacts. *Note: governance and closing-the-loop content is addressed in the team project, not this lab.*

Week-by-Week Schedule

Reading = chapter sections due *before* the listed class session. All chapter titles refer to *Engineering the AI Enterprise*, Parts 3 & 4.

Week 1 Thu Sep 3 (single session – first day)

Date	Topic	Reading Due
Thu Sep 3	Course introduction. What it means to engineer an AI enterprise. The Build & Operate arc. NorthStar Retail introduced – the case that runs the entire semester.	None

📌 Lab 1 Assigned – Thu Sep 3 📌 Due Sat Sep 19, midnight (Lab 1 is a platform setup lab – students can begin AWS environment configuration immediately. The Week 2 platform lectures support completion of the lab.)

Week 2 – AI Systems Development Lifecycle + AI Platform I Tue Sep 8 | Thu Sep 10

Date	Topic	Reading Due
Tue Sep 8	AI Systems Development Lifecycle (AISDLC) – the unifying framework for the course. Why AI development is different. Lifecycle phases, stage gates, and artifacts. How the AISDLC maps to real enterprise practice.	<i>AI Systems Development Lifecycle</i> – Why AI Development Is Different; The AISDLC at a Glance; Lifecycle Phases; Stage Gates and Artifacts
Thu Sep 10	AI Platform & Cloud Architecture I – What is an AI platform? Reference architectures for enterprise AI. Core platform components. The compound-returns argument for platform investment.	<i>AI Platform & Cloud Architecture</i> – Motivation through Core Platform Components

Week 3 – AI Platform II + Data & Feature Engineering I Tue Sep 15 | Thu Sep 17

Date	Topic	Reading Due
Tue Sep 15	AI Platform & Cloud Architecture II – AWS infrastructure for AI. SageMaker domain and ecosystem. Infrastructure as Code with Terraform. Modularity, vendor strategy, and cost governance during build. NorthStar platform architecture walkthrough.	<i>AI Platform & Cloud Architecture</i> – AWS Infrastructure through Key Takeaways

Date	Topic	Reading Due
Thu Sep 17	Data & Feature Engineering I – Why data engineering for AI is different. Three ingestion patterns. Data transformation and operational quality. The Zillow Offers cautionary tale. AWS reference architecture.	<i>Data & Feature Engineering</i> – Motivation through Feature Engineering

📅 **Lab 2 Assigned – Thu Sep 17** 📅 **Due Sat Oct 3, midnight**

Week 4 – Data & Feature Engineering II + Model Development I Tue Sep 22 | Thu Sep 24

Date	Topic	Reading Due
Tue Sep 22	Data & Feature Engineering II – Feature stores and training/serving skew. Data lineage and provenance. Governance, privacy, and security. Airbnb Zipline case study.	<i>Data & Feature Engineering</i> – Feature Stores through Key Takeaways
Thu Sep 24	Model Development I – The development spectrum: choosing your approach. Prompt engineering as an engineering discipline. Training custom models. Fine-tuning foundation models. Reproducibility and model versioning.	<i>Model Development</i> – Motivation through Fine-Tuning Foundation Models; Reproducibility and Model Versioning

Week 5 – Model Development II & III Tue Sep 29 | Thu Oct 1

Date	Topic	Reading Due
Tue Sep 29	Model Development II – RAG – Retrieval-Augmented Generation architecture. Chunking strategies, embedding, reranking. Evaluation of RAG pipelines. When RAG beats fine-tuning and when it doesn't.	<i>Model Development</i> – Retrieval-Augmented Generation section
Thu Oct 1	Model Development III – Agents – Agent design and orchestration. Tool use, memory, planning loops. Failure modes unique to agentic systems. When to use agents vs. simpler approaches. Morgan Stanley AI at Scale case study.	<i>Model Development</i> – Agent Design and Orchestration; AWS Architecture; Key Takeaways

📌 **Lab 3 Assigned – Thu Oct 1 📌 Due Sat Oct 17, midnight**

Week 6 – XOps I & II Tue Oct 6 | Thu Oct 8

Date	Topic	Reading Due
Tue Oct 6	XOps I – DataOps & MLOps – XOps as the operational foundation of enterprise AI. DataOps: building the data foundation. MLOps: automating the model lifecycle. Where most teams fall short.	<i>The XOps Stack</i> – Motivation through MLOps

Date	Topic	Reading Due
Thu Oct 8	XOps II – LLMOps & AgentOps – Operational discipline for foundation model systems. Governance for systems that act. AWS reference XOps architecture. Building XOps capability in your organization.	<i>The XOps Stack</i> – LLMOps through Key Takeaways

Week 7 – Testing & Evaluation Tue Oct 13 | Thu Oct 15

Date	Topic	Reading Due
Tue Oct 13	Testing & Evaluation I – The testing hierarchy for AI systems. Evaluating predictive models: metrics, baselines, holdout strategy. Production readiness criteria. Google Model Cards case study.	<i>Testing and Evaluation</i> – Motivation through Evaluating Predictive Models; Production Readiness Criteria
Thu Oct 15	Testing & Evaluation II – Evaluating generative AI systems. Evaluating agent systems. Fairness, robustness, and safety evaluation. What “passing” a test actually means at enterprise scale.	<i>Testing and Evaluation</i> – Evaluating Generative AI through Key Takeaways

📌 **Lab 4 Assigned – Thu Oct 15** 📌 **Due Sat Oct 31, midnight**

Week 8 – Continuous Delivery Tue Oct 20 | Thu Oct 22

Date	Topic	Reading Due
Tue Oct 20	Continuous Delivery I – CI/CD for AI: what is genuinely different. Continuous integration for AI: testing data, models, and infrastructure in the pipeline. Spotify Hendrix case study.	<i>Continuous Delivery</i> – Motivation through Continuous Integration
Thu Oct 22	Continuous Delivery II – Continuous delivery for AI. Deployment strategies. Infrastructure automation. Pipeline health and measurement.	<i>Continuous Delivery</i> – Continuous Delivery through Key Takeaways

Week 9 – Deployment & Scaling Tue Oct 27 | Thu Oct 29

Date	Topic	Reading Due
Tue Oct 27	Deployment & Scaling I – Deployment strategies: canary, blue/green, shadow, feature flags. Serving infrastructure at scale. Lyft ML Platform case study.	<i>Deployment and Scaling</i> – Motivation through Serving Infrastructure
Thu Oct 29	Deployment & Scaling II – Scaling AI workloads from pilot to full production. Resilience and reliability at scale. Organizational readiness and the operational handoff.	<i>Deployment and Scaling</i> – Scaling AI Workloads through Key Takeaways

📅 Lab 5 Assigned – Thu Oct 29 📅 Due Sat Nov 14, midnight

Week 10 – Security, Privacy & Compliance Tue Nov 3 | Thu Nov 5

Date	Topic	Reading Due
Tue Nov 3	Security, Privacy & Compliance I – The AI security surface. Data security. AI-specific threats: prompt injection, model inversion, adversarial inputs. Security gates in the development lifecycle. AWS infrastructure security.	<i>Security, Privacy & Compliance</i> – Motivation through AI-Specific Threats; Infrastructure Security
Thu Nov 5	Security, Privacy & Compliance II – Privacy engineering. Regulatory compliance frameworks (GDPR, CCPA, EU AI Act). Bridge: what Build hands to Operate. The mindset shift from construction to stewardship.	<i>Security, Privacy & Compliance</i> – Privacy Engineering through Key Takeaways

Week 11 – Metrics, Benchmarks & Guardrails + Monitoring & Observability Tue Nov 10 | Thu Nov 12

Date	Topic	Reading Due
Tue Nov 10	Metrics, Benchmarks & Guardrails – The four performance dimensions. Leading vs. lagging indicators. Guardrail metrics: encoding risk tolerance as numbers. Experimentation and online validation. Stakeholder views.	<i>Metrics, Benchmarks & Guardrails</i> – full chapter

Date	Topic	Reading Due
Thu Nov 12	Monitoring, Observability & Model Lifecycle Management – What monitoring covers in an AI system. Drift detection: the core monitoring problem. Alerting architecture and escalation paths. Observability for generative AI and agent systems. Model lifecycle management.	<i>Monitoring, Observability & Model Lifecycle Management</i> – full chapter

📌 **Lab 6 Assigned – Thu Nov 12 📌 Due Sat Nov 28, midnight**

Week 12 – Reliability Engineering + AI Economics Tue Nov 17 | Thu Nov 19

Date	Topic	Reading Due
Tue Nov 17	Reliability Engineering – SRE principles applied to AI systems. AI-specific failure modes. AWS reliability architecture. Incident response process. Runbook design and rollback procedures.	<i>Reliability Engineering</i> – full chapter
Thu Nov 19	AI Economics – The full cost structure of enterprise AI. Cost estimation as a design discipline. FinOps for AI. Build vs. buy vs. subscribe. Cost-performance tradeoffs and governance.	<i>AI Economics</i> – full chapter

📌 **Lab 7 Assigned – Thu Nov 19 📌 Due Tue Dec 1, midnight** (exception to Saturday convention)

Week 13 – Measuring Business Value + Team Project Launch Tue Nov 24 | ~~Thu Nov 26~~
NO CLASS (Thanksgiving)

Date	Topic	Reading Due
Tue Nov 24	Measuring Business Value – The Metric Pyramid: the central discipline of value translation. The four value dimensions. Attribution: the hard problem. Measuring value for LLM and agentic systems. Shared scorecards. Communicating AI value to non-technical stakeholders. Team project introduced – teams finalized this week.	<i>Measuring Business Value</i> – full chapter
Thu Nov 26	<i>No class – Thanksgiving</i>	–

TEAM PROJECT SESSIONS

Lab 7 is due Tue Dec 1. All remaining class sessions (Weeks 14–15) are devoted to team project work. Chapters on AI Governance and Closing the Loop are assigned as readings; content is discussed in workshop format as it connects to project work.

Week 14 – Team Project Workshop I Tue Dec 1 | Thu Dec 3

Date	Topic	Reading Due
Tue Dec 1	Project Workshop – Team check-ins. Architecture and design review. Q&A on AI Governance content.	<i>AI Governance</i> – full chapter (read before class)
Thu Dec 3	Project Workshop – Team working session. Instructor and TA available for consults. Q&A on Closing the Loop content.	<i>Closing the Loop</i> – full chapter (read before class)

Week 15 – Team Project Workshop II + Final Thoughts Tue Dec 8 | Thu Dec 10 (*last week of class*)

Date	Topic	Reading Due
Tue Dec 8	Project Workshop – Final working session. Team presentations dry run (optional). Open Q&A.	None
Thu Dec 10	Final Thoughts – Where AI engineering goes next. What separates the engineers who shape this from those who follow it. How to keep learning after this course ends.	None

Final Project

Team size: 3-4 students

Due: Thu Dec 17 (last day of finals), 11:59 PM

Teams finalized: TBD, by mid-term.

Prompt: Design a production AI system for a company and use case of your choosing. Your deliverable is a technical design document covering all course layers: platform architecture, data and feature pipeline, model development approach, XOps plan, deployment strategy, operating model (monitoring + reliability), economic justification, and governance framework.

Presentations: During finals week. Each team presents for 15 minutes + 5 minutes Q&A. Schedule posted on Canvas by Dec 1.

Project Grading Rubric:

Dimension	Weight
Technical depth and correctness	40%
Integration and coherence across all layers	30%
Business/executive communication quality	20%
Presentation	10%

Course Policies

Attendance: Not graded, but this course moves fast. We will need to make some changes on the fly. If you have to miss, make sure you check with classmates about content.

AI Tools: You may use AI coding assistants (GitHub Copilot, Claude, etc.) for all work. You must understand and be able to explain everything you submit. If you cannot explain it in office hours, you may not get credit for it.

Academic Honesty: BYU Honor Code applies. Working together on Labs is highly encouraged. Copying lab solutions from other students before the Saturday midnight due date is academic dishonesty.

Office Hours: Tu and Thur after class. It is preferred is you can email for appointments.

Syllabus subject to change. All updates announced on Canvas.